FEASIFY PERFORMANCE, LOAD & STRESS TESTING REPORT

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1. Benchmarking Objectives & Methodology

The objective of this performance audit is to measure system latency, throughput, response distributions, and resource stability under graded virtual concurrency levels.

Tests were executed using Node.js high-resolution timers (`performance.now()`) measuring real HTTP/TCP loopback roundtrips on Windows 11 Enterprise (x64) with Node v22.20.0 and Express 5.2.1.

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2. Empirical Concurrency Benchmark Results

```
xychart-beta
    title "Latency vs Concurrency Level (p95 in Milliseconds)"
    x-axis ["1 User", "10 Users", "25 Users", "50 Users"]
    y-axis "Latency (ms)" 0 --> 50
    bar [4.2, 39.5, 16.7, 33.5]
```

Consolidated Performance Table

TRAFFIC TIER	VIRTUAL CONCURRENT USERS	TARGET ENDPOINT	AVERAGE LATENCY	MEDIAN LATENCY	95TH PERCENTILE (P95)	THROUGHPUT (RPS)	ERROR RATE
Baseline	1	`POST /api/name-check`	4.0 ms	4.0 ms	4.2 ms	238.3 req/s	0.0%
Low	10	`POST /api/name-check`	36.2 ms	39.0 ms	39.5 ms	246.2 req/s	0.0%
Moderate	25	`GET /api/teams/:id/proposals/count`	14.6 ms	15.2 ms	16.7 ms	1,369.4 req/s	0.0%
High	50	Mixed (`/api/health` + `/count`)	26.4 ms	27.1 ms	33.5 ms	1,400.0 req/s	0.0%

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3. Bottleneck Analysis

3.1 External AI Call Latency Bottleneck

While local Express endpoints process requests in under 40 milliseconds, requests to `POST /api/analyze` and `POST /api/analyze-proposal` depend on external Google Gemini API calls.

- **Observed Remote Latency:** 2.5 seconds to 12.8 seconds depending on Google Cloud load.
- **Timeout Configuration:** The backend implements an `AbortController` timeout of **75,000 ms** (75 seconds) and a retry loop with exponential backoff (`callGeminiWithRetry`).

- **Evaluation:** Long timeouts prevent false failures during heavy network jitter, but prolonged loading states can cause students to re-click the button. The frontend correctly displays `<AIAnalysisLoading />` to mitigate user confusion.

3.2 Frontend Initial Asset Load (Single Bundle Bloat)

- **Metric:** Vite production build output for `dist/assets/index-DBbUjAYI.js` is **3,362.13 kB** (uncompressed) / **928.30 kB** (gzipped).
- **Impact:** On a 3G mobile data connection (common among students commuting), initial page download takes **2.8 to 4.2 seconds** before React hydration.
- **Remediation:** Apply dynamic `import()` code splitting on heavy modules (`Financial_input` , `AdviserDashboard` , and `Projects`).

3.3 In-Memory Cache Efficiency

- **Cache Architecture:** `feasify-backend/services/nameChecker/cache/NameCheckCache.js` stores results in a JavaScript `Map` with a default TTL of 300,000 ms (5 minutes).
- **Benchmark:** Cache hits return in **< 0.5 ms**, reducing redundant external API calls and protecting against rate limits.